

Hannah Friedman

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Department of Mathematics
University of California, Berkeley

EMPLOYMENT

Institute for Computational and Experimental Research in Mathematics

September 2026 – May 2027

Institute Postdoctoral Fellow

EDUCATION

University of California, Berkeley

May 2026

PhD in Mathematics

Jointly advised by Bernd Sturmfels and Serkan Hoşten

Thesis: *Metric Algebraic Geometry of Grassmannians*

Harvey Mudd College

May 2023

BS in Mathematics with High Distinction, Honors in Mathematics

Advised by Michael Orrison

Thesis: *Long Increasing Subsequences*

AWARDS & HONORS

Chancellor's Fellowship, UC Berkeley

2023

Alvin White Prize, Harvey Mudd College

2023

Presented to a student or students who have contributed greatly to the humanistic side of the Harvey Mudd College Mathematics Community.

Giovanni Borrelli Mathematics Fellowship, Harvey Mudd College

2021 - 2022

Conferred on one or two junior mathematics majors in recognition of the ability to complete high quality research independently or in tandem with a faculty advisor.

Courtney S. Coleman Prize, Harvey Mudd College

2021

Awarded to three rising juniors who excel in mathematics.

RESEARCH ACTIVITIES

- Visit to Max Planck Institute for Mathematics in the Sciences Leipzig, Summers 2023-2026
- Geometric Methods in Optimization and Numerical Analysis MFO Oberwolfach, July, 2026
- Visit to Academy of Mathematics and Systems Science, CAS Beijing, July 2026
- Organized Metric Algebraic Geometry: Starting Local MPI MiS, April 2026
- Combinatorial Coworkspace Kleinwalsertal, March 2026
- SL Math Summer School: New Perspectives on Discriminants and Applications MPI MiS, June 2025
- Problem-Solving Workshop: Computational Geometric Analysis CUNY Graduate Center, May 2025
- Workshop on the Applications of Commutative Algebra The Fields Institute, May 2025
- Organized Non-Linear Algebra Seminar UC Berkeley, December 2024
- Workshop for Women in Algebraic Statistics St John's College, Oxford, July 2024

PUBLICATIONS

13. Hannah Friedman, Serkan Hoşten, Andrea Rosana. *Distance Optimization Within the Grassmannian*. Submitted (2026).
12. Hannah Friedman. *Maximum Likelihood Estimation on the Grassmannian of Lines*. Submitted (2026).
11. Hannah Friedman, Andrea Rosana, Bernd Sturmfels. *Distance Optimization in the Grassmannian of Lines*. Submitted (2026).

10. Viktoriia Borovik, Hannah Friedman, Serkan Hoşten, Max Pfeffer. *Numerical Algebraic Geometry for Energy Computations on Tensor Train Varieties*. Submitted (2025).
9. Renato G. Bettiol, Hannah Friedman. *Counting Homogeneous Einstein Metrics*. Submitted (2025).
8. Paul Breiding, Taylor Brysiewicz, Hannah Friedman. *Homotopy Iterators*. Submitted (2025).
7. Hannah Friedman, Bernd Sturmfels, Max Wiesmann. *Squared Linear Models*. SIAM Journal on Applied Algebra and Geometry **10** (2026) 153–173.
6. Hannah Friedman, Serkan Hoşten. *Grassmann and Flag Varieties in Linear Algebra, Optimization, and Statistics: An Algebraic Perspective*. Advances in Applied Mathematics **180** (2026) 103130.
5. Yulia Alexandr, Kristen Dawson, Hannah Friedman, Fatemeh Mohammadi, Pardis Semnani, Teresa Yu. *Decomposing Conditional Independence Ideals with Hidden Variables*. Advances in Applied Mathematics **174** (2026) 103003.
4. Hannah Friedman. *Likelihood Geometry of the Squared Grassmannian*. Proceedings of the American Mathematical Society **153** (2025) 4463–4474.
3. Karel Devriendt, Hannah Friedman, Bernhard Reinke, Bernd Sturmfels. *The Two Lives of the Grassmannian*. Acta Universitatis Spientiae, Mathematica, **17** (2025) 8.
2. Hannah Friedman, Bernd Sturmfels, Max Zubkov. *Likelihood Geometry of Determinantal Point Processes*. Algebraic Statistics, **15** (2024) 15–25.
1. Hannah Friedman, Amani R. Maina-Kilaas, Julianna Schalkwyk, Hina Ahmed, and Jamie Haddock. *An Interpretable Joint Nonnegative Matrix Factorization-Based Point Cloud Distance Measure*, Proceedings of the Conference on Information Sciences and Systems (CISS), Baltimore, MD, March 2023.

TALKS

Key: ♡ = colloquium; ♣ = invited; ♠ = contributed; ◇ = seminar/lecture.

- Distance Optimization Within the Grassmannian
 ◇ Math and Data Seminar, University of Missouri–Columbia, October 2026.
- Maximum Likelihood Estimation for Statistical Models on the Grassmannian.
 ♣ Special Session on Applied Algebra in Statistics, Optimization, and Machine Learning,
 AMS Southeastern Sectional Meeting, October 2026.
- ♠ Geometric Methods in Optimization and Numerical Analysis, Oberwolfach, July 2026.
- Numerical Algebraic Geometry for Energy Computations on Tensor Train Varieties.
 ♠ Effective Methods in Algebraic Geometry, July 2026.
- The ML Degree of the Grassmannian
 ♠ Metric Algebraic Geometry at Institut Mittag-Leffler, June 2026.
- Metric Algebraic Geometry of the Grassmannian of Lines.
 ◇ Oberseminar at Universität Magdeburg, April 2026.
 ◇ Algebra Seminar at Universität Osnabrück, April 2026.
- Distance Optimization.
 ◇ Metric Algebraic Geometry: Starting Local, April 2026.
- Counting Homogeneous Einstein Metrics.
 ♡ Santa Clara University Mathematics and Computer Science Colloquium, November 11, 2025.
 ♣ San Francisco State University Algebra, Combinatorics, and Geometry Seminar, October 8, 2025.
 ◇ UC Berkeley Seminar on Commutative Algebra and Algebraic Geometry, September 30, 2025.

- Grassmann and Flag Varieties in Linear Algebra, Optimization, and Statistics: An Algebraic Perspective.
 - ♣ New Frontiers of Geometry and Combinatorics in Machine Learning at SIAM AG, July 10, 2025.
- Squared Linear Models
 - ◇ MPI MiS Seminar on Nonlinear Algebra, June 19, 2025.
- Solving Polynomial Equations (Software Demonstration)
 - ♣ Problem Solving Workshop: Computational Geometric Analysis, June 4, 2025.
- Likelihood Geometry of the Squared Grassmannian
 - ♣ Session on Advances in Applied Algebra and Algebraic Statistics at the AWM Research Symposium, March 27, 2025.
 - ♠ AlgStat 2025, March 27, 2025.
 - ♣ AMS Special Session on Algebraic Statistics in Our Changing World at JMM, January 8, 2025.
 - ◇ UC Berkeley Combinatorics Seminar, October, 2024.
 - ♠ Workshop for Women in Algebraic Statistics at St. John's College, Oxford, July, 2024.
- Likelihood Geometry of the Squared Grassmannian
 - ◇ University of Wisconsin Applied Algebra Seminar, May 15, 2025.
- The Squared Grassmannian
 - ♣ University of Washington Combinatorics and Geometry Seminar, January 15, 2025.
- What is algebraic statistics?
 - ♠ Bay Area Discrete Math Day, December 7, 2024.
- Optimization over Two Embeddings of the Grassmannian
 - ♣ Codex seminar, July 2024.
- Likelihood Geometry of Determinantal Point Processes
 - ◇ Algebra-Geometry-Combinatorics Afternoon at SFSU, November 2023.
- Long Increasing Subsequences
 - ♠ Harvey Mudd Thesis Presentation Days.
- A New Basis for k -Local Class Functions
 - ♠ 2023 Joint Math Meetings.

TEACHING & MENTORSHIP

- Unbounded Representation Leader, Fall 2024 – Fall 2025
- Graduate Student Instructor at UC Berkeley:
 - Math 16B: Analytic Geometry and Calculus, Fall 2025
 - Math 250B: Graduate Commutative Algebra, Spring 2025
 - Math H110: Honors Abstract Linear Algebra, Spring 2025
 - Math 55: Discrete Math, Fall 2024
 - Math 1A: Calculus I, Spring 2024
- UC Berkeley Directed Reading Program Mentor, Spring 2024 & Fall 2025
- President of Women in Math at Harvey Mudd College, Fall 2022-Spring 2023

SKILLS

- Programming Languages: Python, MatLab, Macaulay2, Java, C++, R, Julia
- Languages: English (fluent), German (fluent), Mandarin (proficient), Hebrew (proficient)